Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (2): Normal

Generation Category (1): Design

Load Diversity Factor: Global

Constant kVA=130%, Constant Z=130%, Generic=130%, Constant I=130%

	Swing	V-Control	Load	Total
Number of Buses:	1	0	18	19

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	2	0	0	16	0	0	18

Method of Solution:	Adaptive Newton-Raphson Method
Maximum No. of Iteration:	99
Precision of Solution:	0.0001000
System Frequency:	50.00 Hz
Unit System:	Metric
Project Filename:	Eksisting_PB_SLEMAN
Output Filename:	E:\KULIAH\SKRIPSI\Etap\GI_KENTUNGAN_SKRIPSI\SudahPecahBeban2\SudahPecahBeban2.lfr

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus1	150.000	1	100.0	0.0								
Bus2	20.000	1	100.0	0.0								
Bus3	20.000	1	100.0	0.0								
Bus4	20.000	1	100.0	0.0								
Bus5	20.000	1	100.0	0.0			1.216	0.753				
Bus6	20.000	1	100.0	0.0								
Bus7	20.000	1	100.0	0.0								
Bus8	20.000	1	100.0	0.0								
Bus9	20.000	1	100.0	0.0								
Bus10	20.000	1	100.0	0.0								
Bus11	20.000	1	100.0	0.0								
Bus12	20.000	1	100.0	0.0								
Bus13	20.000	1	100.0	0.0			1.430	0.886				
Bus14	20.000	1	100.0	0.0			1.116	0.692				
Bus15	20.000	1	100.0	0.0			4.593	2.847				
Bus16	20.000	1	100.0	0.0								
Bus17	20.000	1	100.0	0.0			1.125	0.697				
Bus18	20.000	1	100.0	0.0								
Bus19	20.000	1	100.0	0.0								
Total Number of Buses: 19					0.000	0.000	9.480	5.875	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
Bus1	150.000	Swing	1	100.0	0.0					
						0.000	0.000			

Line/Cable/Busway Input Data

Line/Cable/Busway	ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)								
			Length						
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable1	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable2	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable3	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable5	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable6	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable7	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable8	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable9	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable10	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable11	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable12	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable13	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable19	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable21	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable23	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable26	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	

Line / Cable / Busway resistances are listed at the specified temperatures.

2-Winding Transformer Input Data

Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	MVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
TRAFO_05	3-Phase	150.000	150.000	20.000	6.75	5.79	0	0	0	0	0	6.7500	YNyn	0.000
TRAFO_06	3-Phase	150.000	150.000	20.000	12.60	5.79	0	0	0	0	0	12.6000	YNyn	0.000

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
TRAFO_05	2W XFMR	Bus1	Bus2	0.77	4.43	4.50	
TRAFO_06	2W XFMR	Bus1	Bus9	1.43	8.28	8.40	
Cable1	Cable	Bus2	Bus3	3.36	8.66	9.28	
Cable2	Cable	Bus3	Bus4	3.36	8.66	9.28	
Cable3	Cable	Bus5	Bus4	3.36	8.66	9.28	
Cable5	Cable	Bus4	Bus6	3.36	8.66	9.28	
Cable6	Cable	Bus6	Bus7	3.36	8.66	9.28	
Cable7	Cable	Bus7	Bus8	3.36	8.66	9.28	
Cable8	Cable	Bus9	Bus10	3.36	8.66	9.28	
Cable9	Cable	Bus10	Bus11	3.36	8.66	9.28	
Cable10	Cable	Bus11	Bus12	3.36	8.66	9.28	
Cable11	Cable	Bus12	Bus13	3.36	8.66	9.28	
Cable12	Cable	Bus12	Bus14	3.36	8.66	9.28	
Cable13	Cable	Bus12	Bus16	3.36	8.66	9.28	
Cable19	Cable	Bus12	Bus15	3.36	8.66	9.28	
Cable21	Cable	Bus16	Bus17	3.36	8.66	9.28	
Cable23	Cable	Bus16	Bus18	3.36	8.66	9.28	
Cable26	Cable	Bus4	Bus19	3.36	8.66	9.28	

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
* Bus1	150.000	100.000	0.0	9.110	5.915	0.000	0.000	Bus2	1.209	0.754	5.5	84.8	
								Bus9	7.901	5.161	36.3	83.7	
Bus2	20.000	99.957	0.0	0.000	0.000	0.000	0.000	Bus3	1.209	0.753	41.1	84.9	
								Bus1	-1.209	-0.753	41.1	84.9	
Bus3	20.000	99.852	-0.1	0.000	0.000	0.000	0.000	Bus2	-1.208	-0.751	41.1	84.9	
								Bus4	1.208	0.751	41.1	84.9	
Bus4	20.000	99.746	-0.1	0.000	0.000	0.000	0.000	Bus3	-1.207	-0.750	41.1	85.0	
								Bus5	1.207	0.750	41.1	85.0	
								Bus6	0.000	0.000	0.0	0.0	
								Bus19	0.000	0.000	0.0	0.0	
Bus5	20.000	99.640	-0.2	0.000	0.000	1.207	0.748	Bus4	-1.207	-0.748	41.1	85.0	
Bus6	20.000	99.746	-0.1	0.000	0.000	0.000	0.000	Bus4	0.000	0.000	0.0	0.0	
								Bus7	0.000	0.000	0.0	0.0	
Bus7	20.000	99.746	-0.1	0.000	0.000	0.000	0.000	Bus6	0.000	0.000	0.0	0.0	
								Bus8	0.000	0.000	0.0	0.0	
Bus8	20.000	99.746	-0.1	0.000	0.000	0.000	0.000	Bus7	0.000	0.000	0.0	0.0	
Bus9	20.000	99.461	-0.3	0.000	0.000	0.000	0.000	Bus10	7.889	5.088	272.4	84.0	
								Bus1	-7.889	-5.088	272.4	84.0	
Bus10	20.000	98.754	-0.6	0.000	0.000	0.000	0.000	Bus9	-7.859	-5.011	272.4	84.3	
								Bus11	7.859	5.011	272.4	84.3	
Bus11	20.000	98.048	-0.9	0.000	0.000	0.000	0.000	Bus10	-7.829	-4.933	272.4	84.6	
								Bus12	7.829	4.933	272.4	84.6	
Bus12	20.000	97.346	-1.2	0.000	0.000	0.000	0.000	Bus11	-7.799	-4.856	272.4	84.9	
								Bus13	1.353	0.840	47.2	85.0	
								Bus14	1.056	0.656	36.9	85.0	
								Bus16	1.063	0.661	37.1	84.9	
								Bus15	4.327	2.700	151.3	84.8	
Bus13	20.000	97.225	-1.3	0.000	0.000	1.352	0.838	Bus12	-1.352	-0.838	47.2	85.0	
Bus14	20.000	97.251	-1.3	0.000	0.000	1.056	0.654	Bus12	-1.056	-0.654	36.9	85.0	
Bus15	20.000	96.957	-1.4	0.000	0.000	4.318	2.676	Bus12	-4.318	-2.676	151.3	85.0	
Bus16	20.000	97.251	-1.3	0.000	0.000	0.000	0.000	Bus12	-1.062	-0.659	37.1	85.0	

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus17	1.062	0.659	37.1	85.0	
								Bus18	0.000	0.000	0.0	0.0	
Bus17	20.000	97.155	-1.3	0.000	0.000	1.062	0.658	Bus16	-1.062	-0.658	37.1	85.0	
Bus18	20.000	97.251	-1.3	0.000	0.000	0.000	0.000	Bus16	0.000	0.000	0.0	0.0	
Bus19	20.000	99.746	-0.1	0.000	0.000	0.000	0.000	Bus4	0.000	0.000	0.0	0.0	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

\# Indicates a bus with a load mismatch of more than 0.1 MVA

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Percent Loading
Bus1	150.000										10.862	83.9	41.8	
Bus2	20.000										1.424	84.9	41.1	
Bus3	20.000										1.423	84.9	41.1	
Bus4	20.000										1.421	85.0	41.1	
Bus5	20.000				1.207	0.748					1.420	85.0	41.1	
Bus6	20.000													
Bus7	20.000													
Bus8	20.000													
Bus9	20.000										9.387	84.0	272.4	
Bus10	20.000										9.320	84.3	272.4	
Bus11	20.000										9.254	84.6	272.4	
Bus12	20.000										9.187	84.9	272.4	
Bus13	20.000				1.352	0.838					1.590	85.0	47.2	
Bus14	20.000				1.056	0.654					1.242	85.0	36.9	
Bus15	20.000				4.318	2.676					5.080	85.0	151.3	
Bus16	20.000										1.250	85.0	37.1	
Bus17	20.000				1.062	0.658					1.249	85.0	37.1	
Bus18	20.000												-	
Bus19	20.000												-	

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).
# Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

Branch Loading Summary Report

CKT / Branch		Busway / Cable & Reactor			Transformer				
						Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	MVA	%	MVA	%
Cable1	Cable	585.00	41.13	7.03					
Cable2	Cable	585.00	41.13	7.03					
Cable3	Cable	585.00	41.13	7.03					
Cable5	Cable	585.00							
Cable6	Cable	585.00							
Cable7	Cable	585.00							
Cable8	Cable	585.00	272.45	46.57					
Cable9	Cable	585.00	272.45	46.57					
Cable10	Cable	585.00	272.45	46.57					
Cable11	Cable	585.00	47.21	8.07					
Cable12	Cable	585.00	36.86	6.30					
Cable13	Cable	585.00	37.12	6.34					
Cable19	Cable	585.00	151.26	25.86					
Cable21	Cable	585.00	37.12	6.34					
Cable23	Cable	585.00	0.00	0.00					
Cable26	Cable	585.00	0.00	0.00					
TRAFO_05	Transformer				150.000	1.425	0.9	1.424	0.9
TRAFO_06	Transformer				150.000	9.438	6.3	9.387	6.3

* Indicates a branch with operating load exceeding the branch capability.

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop
	MW	Mvar	MW	Mvar	kW	kvar	From	To	in Vmag
Cable1	1.209	0.753	-1.208	-0.751	0.7	1.8	100.0	99.9	0.11
Cable10	7.829	4.933	-7.799	-4.856	29.9	77.1	98.0	97.3	0.70
Cable11	1.353	0.840	-1.352	-0.838	0.9	2.3	97.3	97.2	0.12
Cable12	1.056	0.656	-1.056	-0.654	0.5	1.4	97.3	97.3	0.09
Cable13	1.063	0.661	-1.062	-0.659	0.6	1.4	97.3	97.3	0.10
Cable19	4.327	2.700	-4.318	-2.676	9.2	23.8	97.3	97.0	0.39
Cable2	1.208	0.751	-1.207	-0.750	0.7	1.8	99.9	99.7	0.11
Cable21	1.062	0.659	-1.062	-0.658	0.6	1.4	97.3	97.2	0.10
Cable23	0.000	0.000	0.000	0.000			97.3	97.3	
Cable26	0.000	0.000	0.000	0.000			99.7	99.7	
Cable3	1.207	0.750	-1.207	-0.748	0.7	1.8	99.7	99.6	0.11
Cable5	0.000	0.000	0.000	0.000			99.7	99.7	
Cable6	0.000	0.000	0.000	0.000			99.7	99.7	
Cable7	0.000	0.000	0.000	0.000			99.7	99.7	
Cable8	7.889	5.088	-7.859	-5.011	29.9	77.1	99.5	98.8	0.71
Cable9	7.859	5.011	-7.829	-4.933	29.9	77.1	98.8	98.0	0.71
TRAFO_05	1.209	0.754	-1.209	-0.753	0.2	0.9	100.0	100.0	0.04
TRAFO_06	7.901	5.161	-7.889	-5.088	12.7	73.7	100.0	99.5	0.54
					116.5	341.5			

* This Transmission Line includes Series Capacitor.

Alert Summary Report

	% Alert Settings	
	Critical	**Marginal**
Loading		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
Bus Voltage		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
Generator Excitation		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Marginal Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus12	Bus	Under Voltage	20.000	kV	19.469	97.3	3-Phase
Bus13	Bus	Under Voltage	20.000	kV	19.445	97.2	3-Phase
Bus14	Bus	Under Voltage	20.000	kV	19.450	97.3	3-Phase
Bus15	Bus	Under Voltage	20.000	kV	19.391	97.0	3-Phase
Bus16	Bus	Under Voltage	20.000	kV	19.450	97.3	3-Phase
Bus17	Bus	Under Voltage	20.000	kV	19.431	97.2	3-Phase
Bus18	Bus	Under Voltage	20.000	kV	19.450	97.3	3-Phase

SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	MW	Mvar	MVA	% PF
Source (Swing Buses):	9.110	5.915	10.862	83.87 Lagging
Source (Non-Swing Buses):	0.000	0.000	0.000	
Total Demand:	9.110	5.915	10.862	83.87 Lagging
Total Motor Load:	0.000	0.000	0.000	
Total Static Load:	8.994	5.574	10.581	85.00 Lagging
Total Constant I Load:	0.000	0.000	0.000	
Total Generic Load:	0.000	0.000	0.000	
Apparent Losses:	0.117	0.342		
System Mismatch:	0.000	0.000		

Number of Iterations: 3